Performance and label efficiency of traditional deep-learning models and a retina-specific foundation model for ocular and systemic disease detection: a retrospective comparative study

Samantha Min Er Yew, Xiaofeng Lei*, Jocelyn Hui Lin Goh*, Ke Zou*, Yibing Chen, Sahana Srinivasan, Miao-li Chee, Krithi Pushpanathan, Qingshan Hou, Zhi Da Soh, Can Can Xue, Marco Chak Yan Yu, Charumathi Sabanayagam, E Shyong Tai, Xueling Sim, Ya Xing Wang, Jost B Jonas, Vinay Nangia, Gabriel Dawei Yang, An Ran Ran, Carol Y Cheung, Yangqin Feng, Jun Zhou, Rick Siow Mong Goh, Yukun Zhou, Pearse A Keane, Yong Liu†, Ching-Yu Cheng†, Yih Chung Tham†*

Summary

Background RETFound, a self-supervised retina-specific foundation model, has shown potential in downstream tasks, but its performance in comparison with that of traditional deep-learning models remains unclear. We aimed to evaluate RETFound against three supervised deep-learning models (ResNet50, ViT-Base, and SwinV2) pretrained on images from ImageNet for the detection of several ocular (disease-related visual impairment, visually significant cataract, glaucoma, and diabetic retinopathy) and systemic (diabetes, hypertension, and chronic kidney disease) diseases.

Methods In this retrospective comparative study, all models were fine-tuned on different proportions of the training dataset (100%, 50%, and 20%) and on smaller datasets of fixed sizes (100, 200, and 400 images; 250 and 500 images for diabetic retinopathy) for all tasks. The fine-tuned models were tested on internal datasets from the Singapore Epidemiology of Eye Disease (SEED) study (1842–10 474 images) for all diseases except diabetic retinopathy, for which the Asia Pacific Tele-Ophthalmology Society 2019 dataset (1100 images) was used. Each model was also evaluated on external datasets, comprising population-based datasets (from the Beijing Eye Study, the Central India Eye and Medical Study, the Singapore Prospective Study, and the UK Biobank) and open-source datasets (Ocular Disease Recognition-5K [ODIR-5K], PAPILA, Glaucoma Grading from Multi-Modality Images [GAMMA], Indian Diabetic Retinopathy Image Dataset, and Methods to Evaluate Segmentation and Indexing Techniques in the Field of Retinal Ophthalmology-2). The performance of the models was assessed using 2000 pairwise bootstrap iterations of the area under the receiver operating characteristic curve (AUC) and compared using a two-tailed test with Bonferroni correction, with statistical significance defined as $p<0·017$ to account for the three pairwise comparisons between models.

Findings In internal testing, similar performance was observed for traditional models (AUC 0·914 [95% CI 0·899–0·928] to 0·965 [0·952–0·976]) and RETFound (0·938 [0·920–0·954] to 0·966 [0·951–0·978]) in the detection of ocular disease after fine-tuning on full datasets. With smaller datasets, the performance of all models was similar, except in the case of diabetic retinopathy (≤100 images per class) and glaucoma (≤400 images), for which ResNet50 was inferior to RETFound (all $p≤0·0001$), although the performance of SwinV2 remained similar to that of RETFound. Similar patterns were observed with external test sets. As an example for glaucoma detection, after fine-tuning on 400 images, RETFound had an AUC of 0·908 (95% CI 0·888–0·928) when tested on the internal SEED dataset and, for the external datasets, AUCs of 0·835 (0·816–0·855) when tested on ODIR-5K, 0·779 (0·731–0·826) on PAPILA, and 0·990 (0·974–1·000) on GAMMA, performing significantly better than ResNet50 (SEED $p=0·0001$; ODIR-5k $p<0·0001$; PAPILA $p=0·0032$; and GAMMA $p=0·0010$). For systemic diseases, RETFound consistently outperformed traditional deep-learning models in internal testing when fine-tuned on smaller datasets (≤400 images): for example, for the detection of hypertension when fine-tuned on 100 images, the AUC for RETFound in the SEED dataset was 0·705 (0·687–0·723), compared with 0·634 (0·615–0·654) for ResNet50 and 0·648 (0·628–0·667) for SwinV2 (both $p<0·0001$) and 0·657 (0·637–0·676) for ViT-Base ($p=0·0003$). When tested in the external UKBB dataset, RETFound achieved an AUC of 0·622 (95% CI 0·618–0·625) for the detection of hypertension when fine-tuned on 400 images, 0·599 (0·595–0·602) on 200 images, and 0·599 (0·595–0·603) on 100 images, significantly outperforming ResNet50 and SwinV2 when fine-tuned on 400, 200, and 100 images (all $p≤0·0001$) and ViT-Base when fine-tuned on 400 images ($p<0·0001$) and on 200 images ($p=0·0002$).

Interpretation Under this specific study design, the performance of traditional deep-learning models is similar to that of RETFound for ocular disease detection when fine-tuned on large datasets. By contrast, RETFound shows an advantage in the detection of systemic disease when fine-tuned on smaller datasets. These findings offer insights

Lancet Digit Health 2026; 8: 101031

Published Online August 28, 2026
https://doi.org/10.1016/j.landig.2026.101031

*Joint first authors
†Joint last authors

Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore (S M E Yew BSc, J H L Goh BEng, K Zou PhD, S Srinivasan BEng, K Pushpanathan MSc, Q Hou PhD, Prof E S Tai PhD, Prof C-Y Cheng PhD, Y C Tham PhD); **Centre for Innovation and Precision Eye Health, Yong Loo Lin School of Medicine, National University of Singapore, Singapore** (S M E Yew, J H L Goh, K Zou, S Srinivasan, K Pushpanathan, Q Hou, Prof C-Y Cheng, Y C Tham); **Institute of High-Performance Computing, Agency for Science, Technology and Research, Singapore** (X Lei MSc, Y Feng PhD, J Zhou PhD, R S M Goh PhD, Y Liu PhD); **Singapore Eye Research Institute, Singapore National Eye Centre, Singapore** (J H L Goh, Y Chen BEng, S Srinivasan, M-l Chee BSc, Z D Soh PhD, C C Xue PhD, M C Y Yu PhD, Prof C Sabanayagam PhD, Prof J B Jonas PhD, Y Feng, Prof C-Y Cheng, Y C Tham); **Ophthalmology and Visual Science Academic Clinical Program, Duke-NUS Medical School, Singapore** (Prof C Sabanayagam, Prof C-Y Cheng, Y C Tham); **Saw Swee Hock School of Public Health, National University of Singapore and National University Health System, Singapore** (Prof E Shyong Tai, X Sim PhD); **Signature Research**

into the respective merits and limitations of traditional models and foundation models. Future benchmarking on broader datasets is warranted.

Funding Agency for Science, Technology and Research (A*STAR).

Copyright © 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Programme in Cardiovascular & Metabolic Disorders, Duke-NUS Medical School, Singapore (Prof E S Tai); Precision Health Research, Singapore (Prof E S Tai); Beijing Visual Science and Translational Eye Research Institute, Beijing Tsinghua Changgung Hospital, Tsinghua Medicine, Tsinghua University, Beijing, China (Y X Wang MD, Prof J B Jonas); Tsinghua Medicine, Tsinghua University, Beijing, China (Y X Wang); Rothschild Foundation Hospital, Institut Français de Myopie, Paris, France (Prof J B Jonas); New York Eye and Ear Infirmary of Mount Sinai, Icahn School of Medicine at Mount Sinai, New York, NY, USA (Prof J B Jonas); Suraj Eye Institute, Nagpur, India (V Nangia MD); Department of Ophthalmology and Visual Sciences, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China (G D Yang PhD, A R Ran PhD MMed, Prof C Y Cheung PhD); Hawkes Institute, University College London, London, UK (Y Zhou PhD); NIHR Biomedical Research Centre, Moorfields Eye Hospital NHS Foundation Trust, London, UK (Y Zhou, Prof P A Keane MD); Department of Medical Physics and Biomedical Engineering, University College London, London, UK (Y Zhou); Institute of Ophthalmology, University College London, London, UK (Prof P A Keane)

Correspondence to:
Prof Ching-Yu Cheng, Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, 117549 Singapore
chingyu.cheng@nus.edu.sg

or

Dr Yih Chung Tham, Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, 117549 Singapore
thamyc@nus.edu.sg

Introduction

Artificial intelligence foundation models have emerged rapidly, with notable use cases across several medical domains including cardiology,[1] oncology,[2] histopathology,[3] and ophthalmology.[4–6] The core concept of foundation models involves pretraining deep-learning models on vast amounts of unlabelled medical data through pretext tasks, often using a self-supervised learning approach.[7] These foundation models are then fine-tuned on minimal domain-specific labelled data for downstream tasks. By contrast, approaches to traditional deep-learning models typically involve pretraining on natural images (eg, from ImageNet) using supervised learning, and then further fine-tuning on domain-specific labelled data for downstream tasks.[8]

RETFound, the first foundation model in ophthalmology, was released in 2023.[4] This retina-specific model was pretrained on 1·3 million natural images followed by around 900 000 colour fundus photographs (CFPs) and around 700 000 optical coherence tomography (OCT) scans using self-supervised learning (ie, masked autoencoders). After subsequent fine-tuning, RETFound showed promising performance when tested on several downstream tasks, including the diagnosis of retinal disease and the prediction of systemic disease.[4]

Although the original RETFound study[4] showed the promising potential of a foundation model-based pretraining approach, several knowledge gaps remain. First, the comparative performance and utility of the fine-tuning approach of RETFound relative to current, frequently used traditional deep-learning models that use supervised learning approaches remains incompletely understood. Second, comparisons between these models in terms of crucial aspects such as label efficiency (ie, performance relative to the amount of fine-tuned data), computational resources, and generalisability have not been thoroughly explored. A more holistic evaluation is therefore needed, encompassing a broad range of downstream tasks with extensive external testing.

To address these gaps, we aimed to compare the performance of RETFound with that of models trained using traditional deep-learning approaches on ocular and systemic disease detection tasks. The residual network model ResNet50[9] and the vision transformer-based model ViT-Base[10] were selected because of their widespread use,[11–15] whereas SwinV2,[16] a model based on the Swin transformer with architectural advantages and strong performance in previous benchmark studies,[11,15] was included to provide a more comprehensive evaluation and to broaden the scope of comparison. This study aimed to provide insights into the key considerations for adopting RETFound versus traditional deep-learning models, including performance, label efficiency, and computational demands.

Research in context

Evidence before this study

We searched PubMed for articles published in English from database inception to Jul 28, 2026, using the search terms "RETFound" and "vision transformer" and "convolutional neural network". We found 24 studies that evaluated RETFound against conventional deep-learning models (ie, vision transformer and/or convolutional neural networks) using coloured retinal photographs for ocular or systemic disease detection. However, the majority of the studies (n=21) focused primarily on fine-tuning RETFound for single disease outcomes, were not externally validated, or both. To the best of our knowledge, no previous single study has reported a comprehensive comparative evaluation of RETFound and traditional deep-learning models across major ocular and systemic disease outcomes, including the use of multiple datasets and reducing the size of the fine-tuning dataset to a fixed proportion of the full dataset or to a fixed number of images.

Added value of this study

Our study provides a comprehensive, head-to-head evaluation of model performance and label efficiency, across multiple internal and external test sets derived from diverse population-based studies and open-source datasets, for various ocular and systemic disease outcomes. Our findings contribute to evidence on the performance advantages of RETFound over traditional deep-learning approaches, highlighting which models show better or similar performance in specific scenarios.

Implications of all the available evidence

Our study highlights the continued usefulness of traditional deep-learning models for ocular disease detection when large datasets are available for fine-tuning. By contrast, RETFound has advantages for systemic disease detection in data-scarce settings. However, as deep-learning techniques continue to evolve and new foundation models emerge, further benchmarking evaluations will be essential.

Methods

Study design

An overview of this retrospective comparative study is shown in figure 1. We fine-tuned RETFound and three traditional deep-learning models (ResNet50,[9] SwinV2,[16] and ViT-Base[10]) using various quantities of retinal images to detect four ocular diseases (disease-related visual impairment, visually significant cataract, diabetic retinopathy, and glaucoma) and three systemic diseases (diabetes, hypertension, and chronic kidney disease). Disease definitions are provided in the appendix (p 2). Models were tested internally, on data from the same source as the fine-tuning set, and externally, on data from independent sources not seen during fine-tuning. Characteristics of each dataset—including the number of images used for development (ie, fine-tuning), internal testing, and external testing for each task and the baseline characteristics of the participants, including sex and ethnicity data—are summarised in the appendix (pp 3–4).

Written informed consent was obtained from all participants before their enrolment in the original studies from which the retinal image datasets were obtained. Each study complied with the principles of the Declaration of Helsinki and received approval from the respective local ethical committees. Permission for the use of data was granted by the principal investigator of each study.

Datasets for ocular disease detection

For the detection of disease-related visual impairment, the internal dataset was from the Singapore Epidemiology of Eye Disease (SEED) study, with 333 (8·8%) of 3803 eyes analysed classed as positive cases. This dataset comprised participants from the Singapore Malay Eye Study (SIMES), the Singapore Chinese Eye Study (SCES), and the Singapore Indian Eye Study (SINDI).[17] External datasets were from the Central India Eye and Medical Study (CIEMS),[18] with 662 (10·1%) of 6526 cases classed as positive; the Beijing Eye Study (BES),[19] with 384 (6·2%) of 6239 cases positive; and the Chinese University of Hong Kong's Sight Threatening Diabetic Retinopathy study (CUHK-STDR),[20] with 136 (14·0%) of 968 cases positive.

Visually significant cataract refers to eyes with cataract (ie, nuclear sclerosis, cortical, and posterior subcapsular cataract) and concurrent best corrected visual acuity of worse than 20/60 (log [minimum angle of resolution] >0·48). For the detection of visually significant cataract, we used the SIMES dataset as the internal dataset, with 74 (4·0%) of 1849 eyes analysed classed as positive cases. External datasets were from SCES, with 144 (2·2%) of 6409 cases classed as positive; SINDI, with 146 (2·3%) of 6411 cases positive; and BES, with 77 (1·6%) of 4881 cases positive.

Classification of diabetic retinopathy was based on the International Clinical Diabetic Retinopathy severity scale,[21] which has five classes: no diabetic retinopathy, mild non-proliferative diabetic retinopathy, moderate non-proliferative diabetic retinopathy, severe non-proliferative diabetic retinopathy, and proliferative diabetic retinopathy. For the detection of diabetic retinopathy, we used the Asia Pacific Tele-Ophthalmology Society 2019 (APTOS-2019)[22] dataset as the internal dataset. Of 1100 eyes analysed, 542 (49·3%) had no diabetic retinopathy, 111 (10·1%) had mild non-proliferative diabetic retinopathy, 300 (27·3%) had moderate non-proliferative diabetic retinopathy, 58 (5·3%) had severe non-proliferative diabetic retinopathy, and 89 (8·1%) had proliferative diabetic retinopathy. The external datasets were the Indian Diabetic Retinopathy Image Dataset (IDRiD),[23] in which, of 516 eyes, 168 (32·6%) had no diabetic retinopathy, 25 (4·8%) had mild non-proliferative diabetic retinopathy, 168 (32·6%) had moderate non-proliferative diabetic retinopathy, 93 (18·0%) had severe non-proliferative diabetic retinopathy, and 62 (12·0%) had proliferative diabetic retinopathy; and the Methods to Evaluate Segmentation and Indexing Techniques in the Field of Retinal Ophthalmology-2 (Messidor-2) dataset,[24] in which, of 1744 eyes, 1017 (58·3%) had no diabetic retinopathy, 270 (15·5%) had mild non-proliferative diabetic retinopathy, 347 (19·9%) had moderate non-proliferative diabetic retinopathy, 75 (4·3%) had severe non-proliferative diabetic retinopathy, and 35 (2·0%) had proliferative diabetic retinopathy.

See Online for appendix

For the detection of glaucoma, we used the SEED dataset as the internal dataset, with 137 (1·3%) of 10 474 eyes analysed classed as positive cases. External datasets were Ocular Disease Recognition-5K (ODIR-5K),[25] with 397 (6·2%) of 6392 cases classed as positive; PAPILA,[26] with 155 (31·8%) of 488 cases positive; and Glaucoma Grading from Multi-Modality Images (GAMMA),[27] with 50 (50·0%) of 100 cases positive.

Datasets for systemic disease detection

For the detection of systemic disease, we used the SEED dataset as the internal dataset. In this dataset, 1799 (27·7%) of 6484 eyes analysed were classed as positive for diabetes, 1961 (60·7%) of 3233 as positive for hypertension, and 221 (12·0%) of 1842 as positive for chronic kidney diseases. External datasets were from BES (632 [12·1%] of 5244 positive for diabetes, 2425 [50·2%] of 4835 positive for hypertension, and 84 [2·9%] of 2937 positive for chronic kidney disease); the Singapore Prospective Study[28,29] (SP2; 1346 [17·7%] of 7620 positive for diabetes, 3140 [41·2%] of 7620 positive for hypertension, and 443 [5·7%] of 7714 positive for chronic kidney disease); and the UK Biobank[30] (UKBB; 4064 [5·4%] of 75 220 positive for diabetes, 52 349 [55·2%] of 94 912 positive for hypertension, and 2325 [2·2%] of 104 097 positive for chronic kidney disease).

We used macular-centred CFPs for all tasks and included both macular-centred CFPs and optic disc photographs for the detection of glaucoma, diabetes, and hypertension. Standard field of view images were used (range 30–50°). Poor-quality images from the SEED (ie, from SIMES, SCES, and SINDI), SP2, and UKBB datasets were excluded using our in-house algorithm.[31] Image quality in the BES, CIEMS, and CUHK-STDR datasets was assessed through

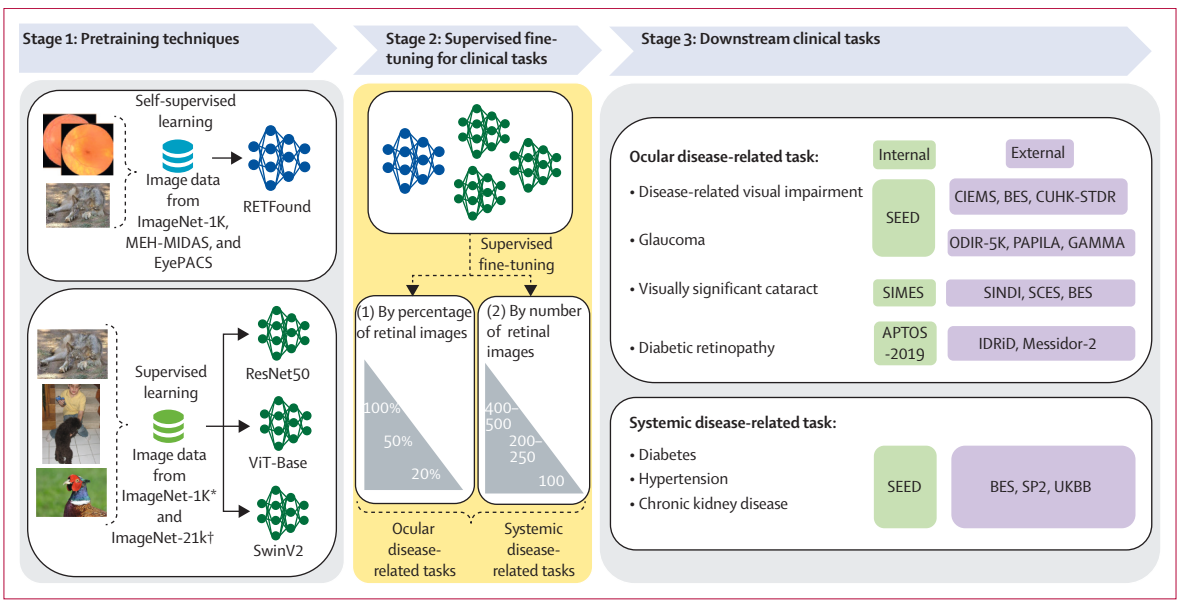


Figure 1: **Overview of study design**
APTOS-2019=Asia Pacific Tele-Ophthalmology Society 2019. BES=Beijing Eye Study. CIEMS=Central India Eye and Medical Study. CUHK-STDR=Chinese University of Hong Kong's Sight Threatening Diabetic Retinopathy study. GAMMA=Glaucoma Grading from Multi-Modality Images. IDRiD=Indian Diabetic Retinopathy Image Dataset. MEH-MIDAS=Moorfields Diabetic Image Dataset. Messidor-2=Methods to Evaluate Segmentation and Indexing Techniques in the Field of Retinal Ophthalmology. ODIR-5K=Ocular Disease Recognition-5K. SCES=Singapore Chinese Eye Study. SEED=Singapore Epidemiology of Eye Disease. SIMES=Singapore Malay Eye Study. SINDI=Singapore Indian Eye Study. SP2=Singapore Prospective Study. UKBB=UK Biobank. *ResNet50 only. †ViT-Base and SwinV2 only.

manual inspection by the respective study representatives. JBJ reviewed the BES and CIEMS datasets and CYC reviewed the CUHK-STDR dataset. Final adjudication was done by SMEY, JHLG, and YCT.

Model pretraining and fine-tuning

The fine-tuning pipelines and backbone architectures of ResNet50, SwinV2, ViT-Base, and RETFound are shown in the appendix (pp 1, 43–44). All models were fine-tuned using two distinct approaches (figure 1). First, we varied the sample sizes used for fine-tuning by using different proportions (ie, 20%, 50%, and 100%) of the full dataset. Second, we used smaller samples containing a fixed number of images, with balanced class ratios of disease-positive and disease-negative cases. Specifically, for binary detection tasks (ie, disease-positive and disease-negative cases), we included 400, 200, and 100 images, and for the classification of diabetic retinopathy into five classes by severity we used 500 and 250 images.

For the traditional deep-learning models, ResNet50 was initialised with pretrained weights derived through supervised learning from ImageNet-1K, containing 1·4 million labelled natural images. ViT-Base and SwinV2 were initialised with pretrained weights derived through supervised learning from ImageNet-21K, consisting of 14 million labelled natural images. Next, we adopted these pretrained models and fine-tuned all layers on task-specific retinal imaging data using supervised learning. This approach, referred to as transfer learning, leverages representations learned from a large source domain (natural images from ImageNet) and adapts them to a different, but related, target domain (retinal images). The original classification head was replaced with a task-specific classifier, and this procedure was applied across all three traditional deep-learning models. Fine-tuning was done for a fixed 40 epochs, which balances sufficient learning (convergence) with the need to avoid overfitting (generalisation). The best checkpoint was selected on the basis of validation performance. Although model performance typically plateaued before 40 epochs and early stopping could have been used, we fixed training at 40 epochs to ensure consistent comparisons across dataset sizes. Fine-tuning parameters included a batch size of 16, a fixed learning rate of 1×10^{-5}, a weight decay of 5×10^{-4}, and a default drop path rate. There was no layer-wise learning rate decay for the traditional models (appendix pp 35–41).

By contrast, RETFound was initialised with pretrained weights derived using self-supervised learning (ie, masked autoencoders) from ImageNet-1K, followed by 904 170 unlabelled CFPs from the Moorfields Diabetic Image Dataset (MEH-MIDAS) and Kaggle EyePACs.[4] We adopted the methods outlined in the original RETFound study,[4] which used full-layer fine-tuning and applied label smoothing to regulate the output distribution and reduce overfitting. We modified only the classification layer to match the target class. Although full-layer fine-tuning could increase the risk of overfitting on smaller datasets, a comparison with freezing the encoder and fine-tuning only the classification head of RETFound showed similar results, indicating that full-layer fine-tuning is appropriate (appendix p 42). Fine-tuning was conducted for 50 epochs

on the 20%, 50%, and 100% fine-tuned datasets, and 100 epochs on the remaining datasets. We used a batch size of 16, a base learning rate of 5×10^{-3}, a weight decay of 0·05, a drop path rate of 0·2, and a layer-wise learning rate decay of 0·65. These configurations enable optimised performance to support the pretraining scheme and architecture of RETFound.[32]

For image preprocessing, we detected the retinal mask using the Hough circle transform,[33] then cropped the mask region. For data augmentation in the traditional deep-learning models, we applied random horizontal flipping, scaling, colour jitter, affine, and rotation after resizing and cropping (to 256×256 pixels for SwinV2 and to 224×224 pixels for ViT-Base and ResNet50). For RETFound, we followed the original published methods,[4] applying the default rand-m9-mstd0.5-inc1 RandAugment policy. This policy applies random transformations such as rotation, shear, and colour distortion with added intensity variability. All fundus images were then cropped to 224×224 pixels.

To address class imbalance in training, we applied the square-root inverse reweighting technique. This method assigns weights to each class on the basis of the inverse square root of its frequency, thereby increasing the importance of under-represented classes without excessively amplifying them.[34,35] This approach is particularly relevant for the larger datasets (ie, when fine-tuning on 100%, 50%, or 20% of the data), which were naturally imbalanced between disease-positive and disease-negative groups. For the smaller, fixed-size subsets (ie, with 200, 100, and 50 images per class), class balance was enforced during subset selection, therefore the effect of square-root inverse reweighting was expected to be negligible. Nevertheless, for methodological consistency across tasks, we applied this technique uniformly.

Overall, these pretraining and fine-tuning strategies were selected because they represent the commonly adopted configurations for each model, emphasising practicality, reproducibility, and relatively fair comparisons. Additionally, training all architectures under a single pretraining dataset would either artificially weaken transformer baselines (in the case of ViT-Base and SwinV2) when limited to ImageNet-1K, or would render training from scratch prohibitively costly (in the case of ResNet50) if the models were retrained and a switch to ImageNet-21K was needed. Furthermore, these chosen configurations have been extensively benchmarked, have shown to be stable and well optimised,[11–15] and are aligned with prevailing implementations in the deep-learning community, including commonly used libraries such as PyTorch Image Models,[36] PyTorch Torchvision,[37] and TensorFlow Model Garden.[38]

Fine-tuning and computational evaluation of all models was conducted using PyTorch (Python 3.8, torch 1.7.0, and Torchvision 0.8.0) on a workstation equipped with an NVIDIA RTX 3090 graphics processing unit (GPU) and an Intel Xeon W-2275 central processing unit with a clock speed of 3·30 GHz, running Ubuntu 20.04.6 long-term support. Further training parameters and computational resource use are shown in the appendix (pp 35–41).

Statistical analysis

For binary detection tasks, we ran 2000 bootstrap iterations of the area under the receiver operating characteristic curve (AUC), maximum F1 score, F1 score, sensitivity, and precision at an 80% specificity threshold.[39] For the five-class diabetic retinopathy detection task, we reported macro-averaged versions of these metrics. 95% CIs were included. Two-tailed test pairwise bootstrap comparisons between RETFound and the three traditional deep-learning models were conducted with Bonferroni correction[40,41] for AUC and maximum F1 score. Statistical significance was defined as $p<0·017$ (0·05/3), accounting for the three pairwise comparisons.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

The demographic characteristics of participants in all study datasets are provided in the appendix (pp 3–4). The SEED dataset (Singapore) includes Malay (SIMES), Indian (SINDI), and Chinese (SCES) participants aged 40·0–91·3 years. The CIEMS (India) dataset comprises Indian participants aged 30–100 years, whereas the BES (China) and CUHK-STDR (Hong Kong) datasets consist of Chinese participants aged 50–93 years and 15·2–86·7 years, respectively. The SP2 dataset (Singapore) includes Chinese, Indian, and Malay individuals aged 24·6–94·9 years, whereas UKBB (UK) includes individuals aged 40–76 years who are White, Asian, of African ancestry, and of other ethnicities. Approximately half of the individuals in each dataset were female. Only country of origin was publicly available for the open-source datasets, each of which were collected at institutions in their respective countries—APTOS-2019 and IDRiD in India, Messidor-2 in France, ODIR-5K and GAMMA in China, and PAPILA in Spain—and individual patient demographics were not publicly available. Representative examples of fundus photographs from each dataset are shown in the appendix (p 45).

For the detection of ocular diseases, similar performance was observed for traditional models (AUC 0·914 [95% CI 0·899–0·928] to 0·965 [0·952–0·976]) and RETFound (0·938 [0·920–0·954] to 0·966 [0·951–0·978]) after fine-tuning on full datasets. Performance remained similar after fine-tuning on 50% and 20% of the images and on external test sets (figures 2, 3A, appendix pp 5–21). When fine-tuned on reduced sample sizes of 100, 200, or 400 images, similar performances between RETFound and the traditional deep-learning models persisted for the detection of disease-related visual impairment and visually significant cataract. For glaucoma detection, RETFound consistently

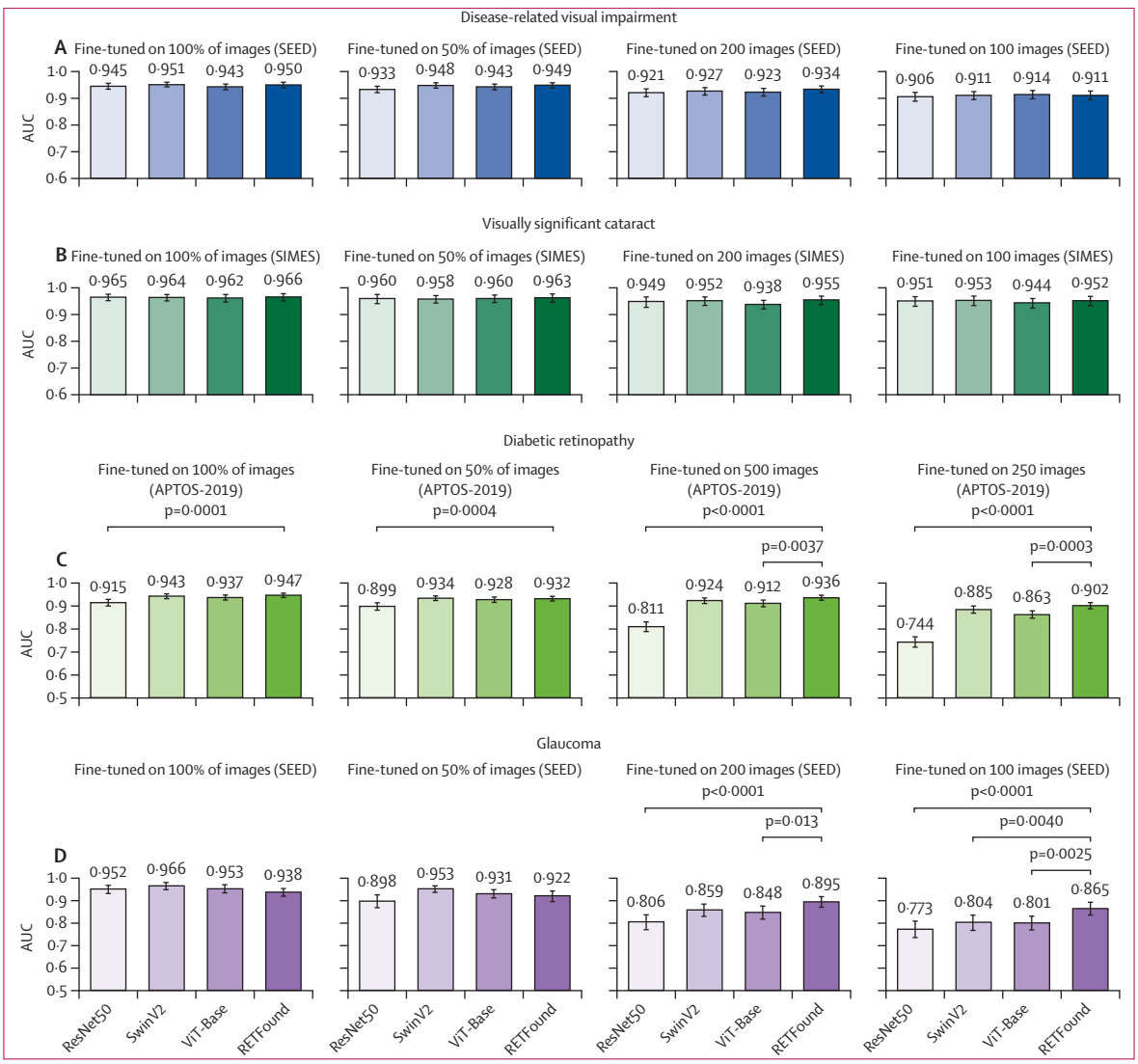


Figure 2: **Performance of the models for ocular disease detection in the internal test set when fine-tuned on various dataset sizes**
(A) Disease-related visual impairment. (B) Visually significant cataract. (C) Diabetic retinopathy. (D) Glaucoma. Error bars are 95% CI. APTOS-2019=Asia Pacific Tele-Ophthalmology Society 2019. AUC=area under the receiver operating characteristic curve. SEED=Singapore Epidemiology of Eye Disease. SIMES=Singapore Malay Eye Study. Detailed pairwise p values are provided in the appendix (p 50).

showed the highest AUC among models. As an example, after fine-tuning on 400 images, RETFound achieved an AUC of 0·908 (95% CI 0·888–0·928) when tested on the internal SEED dataset and, for the external datasets, AUCs of 0·835 (0·816–0·855) when tested on ODIR-5K, 0·779 (0·731–0·826) on PAPILA, and 0·990 (0·974–1·000) on GAMMA. In this setting, RETFound performed significantly better than ResNet50 (SEED p=0·0001; ODIR-5k p<0·0001; PAPILA p=0·0032; and GAMMA p=0·0010; appendix p 15). For the detection of diabetic retinopathy, after fine-tuning RETFound on 500 images, internal testing resulted in an AUC of 0·936 (95% CI 0·925–0·947) with APTOS-2019, with external testing giving AUCs of 0·812 (0·786–0·835) for IDRiD and 0·712 (0·695–0·727) for Messidor-2. In this setting, RETFound performed significantly better than ResNet50 (APTOS-2019 p<0·0001; IDRiD p<0·0001; and Messidor-2 p=0·0019) and ViT-Base (APTOS-2019 p=0·0037; IDRiD p=0·0018; Messidor-2 p=0·0013; appendix pp 20–21).

In both scenarios (fine-tuning on smaller, fixed datasets), RETFound and SwinV2 showed similar performance for the majority of comparisons. The confusion matrices, maximum F1 scores, sensitivity at 80% specificity, F1 scores at 80% specificity, and precision at 80% specificity mirrored these trends (appendix pp 5–21, 46–47). For example, in the detection of disease-related visual impairment, all models performed similarly in terms of sensitivity when fine-tuned on 100% of the available training data (figure 3A, appendix p 46). The numbers of false positives and false negatives produced by each model were,

respectively, 709 and 21 for ResNet50 (sensitivity 93·7%), 695 and 24 for ViT-Base (92·8%), 689 and 20 for SwinV2 (94·0%), and 691 and 21 for RETFound (93·7%).

In the detection of systemic diseases, RETFound generally did not exhibit distinctly superior performance compared with traditional deep-learning models when fine-tuned on large datasets (ie, ≥20% of images; figures 3B, 4, appendix pp 22–34). However, an advantage became more evident with smaller sample sizes (ie, 100, 200, or 400 images).

In the detection of diabetes, after fine-tuning on up to 400 images, RETFound yielded the highest AUC among models in most internal and external datasets (appendix pp 22–34). For example, when tested on the SEED dataset, RETFound achieved an AUC of 0·697 (95% CI 0·683–0·712) when fine-tuned on 400 images, 0·708 (0·694–0·723) on 200 images, and 0·648 (0·632–0·663) on 100 images. RETFound significantly outperformed ResNet50 when fine-tuned on 400 images ($p<0·0001$), 200 images ($p<0·0001$), and 100 images ($p=0·0013$); ViT-Base when fine-tuned on 400 images and 200 images ($p<0·0001$ for both); and SwinV2 when fine-tuned on 100 images ($p<0·0001$).

For the detection of hypertension, when fine-tuned on 100%, 50%, or 20% of images, RETFound did not show significant differences with the traditional deep-learning models in internal tests (figure 4B, appendix pp 26–30). However, when fine-tuned on 100, 200, or 400 images, across both internal and external test sets, RETFound showed had highest AUC among the tested models. For example, in the UKBB dataset, RETFound achieved an AUC of 0·622 (95% CI 0·618–0·625) when fine-tuned on 400 images, 0·599 (0·595–0·602) on 200 images, and 0·599 (0·595–0·603) on 100 images. RETFound significantly outperformed ResNet50 and SwinV2 when fine-tuned on 200 and 100 images (both $p<0·0001$) and on 400 images (ResNet50 $p=0·0001$ and SwinV2 $p<0·0001$) and ViT-Base when fine-tuned on 400 images ($p<0·0001$) and on 200 images ($p=0·0002$). In another example, in the SEED dataset, RETFound achieved an AUC of 0·705 (0·687–0·723) when fine-tuned on 100 images. This performance was significantly better than that of ResNet50 (0·634 [0·615–0·654]) and SwinV2 (0·648 [0·628–0·667]; both $p<0·0001$) and ViT-Base (0·657 [0·637–0·676], $p=0·0003$).

In the detection of chronic kidney disease, when fine-tuned on 100% or 50% of images, no significant differences were observed between RETFound and the traditional deep-learning models in internal tests (figure 4C, appendix pp 31–34). When fine-tuned on 400, 200, or 100 images, RETFound generally showed the best performance among the models tested. After fine-tuning on 100 images, RETFound achieved an AUC of 0·822 (95% CI 0·800–0·843) when tested on the internal SEED dataset and, for the external datasets, AUCs of 0·832 (0·795–0·866) on BES, 0·776 (0·752–0·799) on SP2, and 0·635 (0·625–0·645) on UKBB. In this setting, RETFound significantly outperformed ResNet50 on the SP2 ($p<0·0001$) and UKBB ($p=0·014$) datasets, ViT-Base on all datasets ($p<0·0001$ for SEED, SP2, and UKBB and $p=0·0082$ for BES), and SwinV2 on all datasets ($p<0·0001$ for BES, SP2, and UKBB and $p=0·0054$ for SEED). Furthermore, confusion matrices, maximum F1 score, sensitivity at 80% specificity, F1 scores at 80% specificity, and precision at 80% specificity for systemic disease detection showed that RETFound generally outperformed traditional deep-learning models when fine-tuned on smaller sample sizes of 100, 200, or 400 images (appendix pp 22–34, 48–49). For example, in a systemic disease task such as the detection of diabetes (appendix p 48), the sensitivities of all models when fine-tuned on 100% of images were generally lower than those reported for ocular disease detection (appendix pp 46–47). However, when fine-tuned on a smaller dataset of 400 images, RETFound achieved a higher sensitivity than the other models. The numbers of false positives and false negatives from each model were 937 and 1134 for ResNet50 (sensitivity 37·0%), 938 and 1080 for ViT-Base (40·0%), 938 and 993 for SwinV2 (44·8%), and 935 and 974 for RETFound (45·9%).

During cross-ethnicity evaluation, we observed that all models performed suboptimally when the ethnicity of individuals in the testing dataset did not align with that of the fine-tuning dataset. In the diabetic retinopathy detection task, all models fine-tuned on the APTOS-2019 dataset (comprising data from Indian individuals) showed a larger decrease in AUC (around 20–25%) when validated on Messidor-2 (comprising data from French individuals) than when tested on IDRiD (comprising data from Indian individuals; 12–17%). Across systemic disease detection tasks, all models fine-tuned on SEED (comprising data from Chinese, Malay, and Indian individuals) exhibited greater performance decreases (around 10–20%) when evaluated on the UKBB dataset (featuring data from predominantly White and South Asian individuals) than on datasets more demographically similar to SEED, such as SP2 (<10%; appendix pp 19–34). However, despite the decrease in performance in external tests, RETFound retained a relative advantage when fine-tuned on smaller datasets for systemic diseases and the more complex ocular disease task of diabetic retinopathy and glaucoma classification. In the detection of disease-related visual impairment, models fine-tuned on SEED (comprising data from individuals aged 40·0–85·8 years) showed a greater decrease in AUC (around 5–10%) when externally tested on CUHK-STDR (population aged 15·2–86·7 years with diabetes) than when tested on CIEMS (aged 30–100 years; <5%) and BES (aged 50–93 years; <5%).

During fine-tuning, RETFound had the highest computational demands in terms of model size, GPU memory, and fine-tuning duration. For example, in the task for the detection of visually significant cataract, RETFound had a model size of 303 MB, a GPU memory of 12·3 GB, and a fine-tuning time of 2 min 1 s, compared with 87 MB, 11·4 GB, and 1 min 57 s for SwinV2; 86 MB, 3·4 GB, and

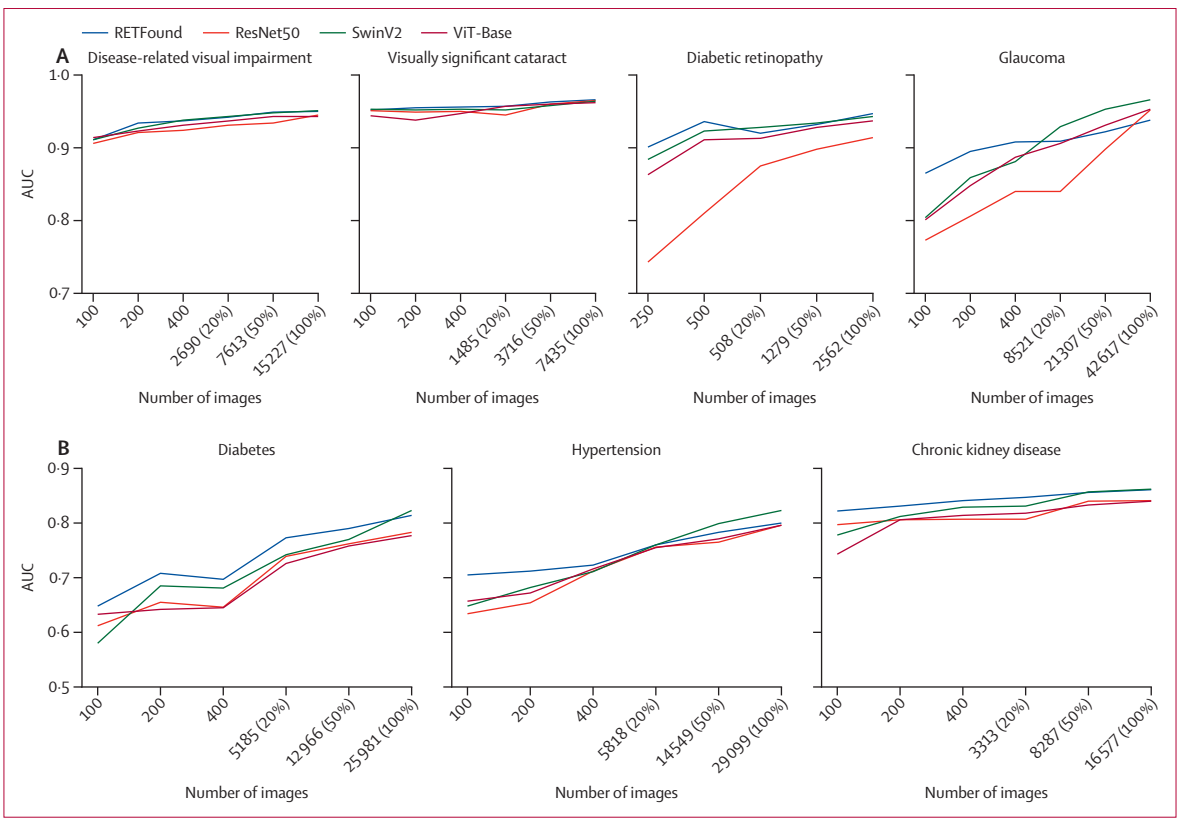


Figure 3: **Performance of each model in the detection of ocular and systemic diseases in the internal test set by number of images in the fine-tuning dataset**
(A) Ocular disease detection. (B) Systemic disease detection. AUC=area under the receiver operating characteristic curve.

59 s for ViT-Base; and 24 MB, 2·3 GB, and 21 s for ResNet50. During inference, RETFound used less GPU memory than did SwinV2 (6582 MB *vs* 9980 MB), but had a slightly longer inference time (82·0 images per s *vs* 93·4 images per s). Both ViT-Base (2662 MB and 149·9 images per s) and ResNet50 (2032 MB and 194·6 images per s) required less GPU memory and had shorter inference times than RETFound. Similar patterns were observed across other tasks (appendix pp 35–41). Additionally, across most tasks, models fine-tuned on imbalanced datasets often achieved higher AUCs than those trained on balanced datasets (appendix pp 5–34).

Discussion

Our study evaluated and compared the performance, label efficiency, and computational resource use of RETFound and traditional deep-learning model approaches (ResNet50, ViT-Base, and SwinV2) across various ocular and systemic disease detection tasks, using various dataset sizes for fine-tuning and multiple external tests. For the detection of ocular diseases, traditional deep-learning models had similar performance to that of RETFound when fine-tuned on large datasets (ie, ≥20% of data), supporting the continued usefulness of such models in these tasks. Conversely, for systemic diseases, RETFound performed better than traditional deep-learning models in most of the comparisons when fine-tuned on smaller datasets (≤400 images). Although the differences in performance were small (AUC difference <0·15), they highlight the potential value of RETFound for oculomics tasks, particularly in settings in which well annotated and well labelled datasets are scarce. However, the higher computational demands of RETFound compared with the tested traditional deep-learning models could limit the feasibility of its deployment in low-resource settings.

The similarity in ocular disease detection performance between RETFound and traditional deep-learning models when fine-tuned on large datasets can probably be attributed to the availability of sufficient data, coupled with the distinctiveness of disease features in retinal images. Signs of ocular disease (eg, retinal haemorrhage) are often more obvious than the subtle retinal features present in systemic diseases (eg, retinal microvascular abnormalities or vessel geometry changes),[42–44] making feature detection for ocular diseases a relatively easier task. As a result, traditional models are able to perform comparably well. Additionally, when fine-tuning sample sizes were reduced (≤500 images), RETFound performed better than ResNet50 in both internal and external test sets for the detection of diabetic retinopathy and glaucoma. By contrast, the performance of SwinV2 was similar to that of RETFound. This observation is consistent

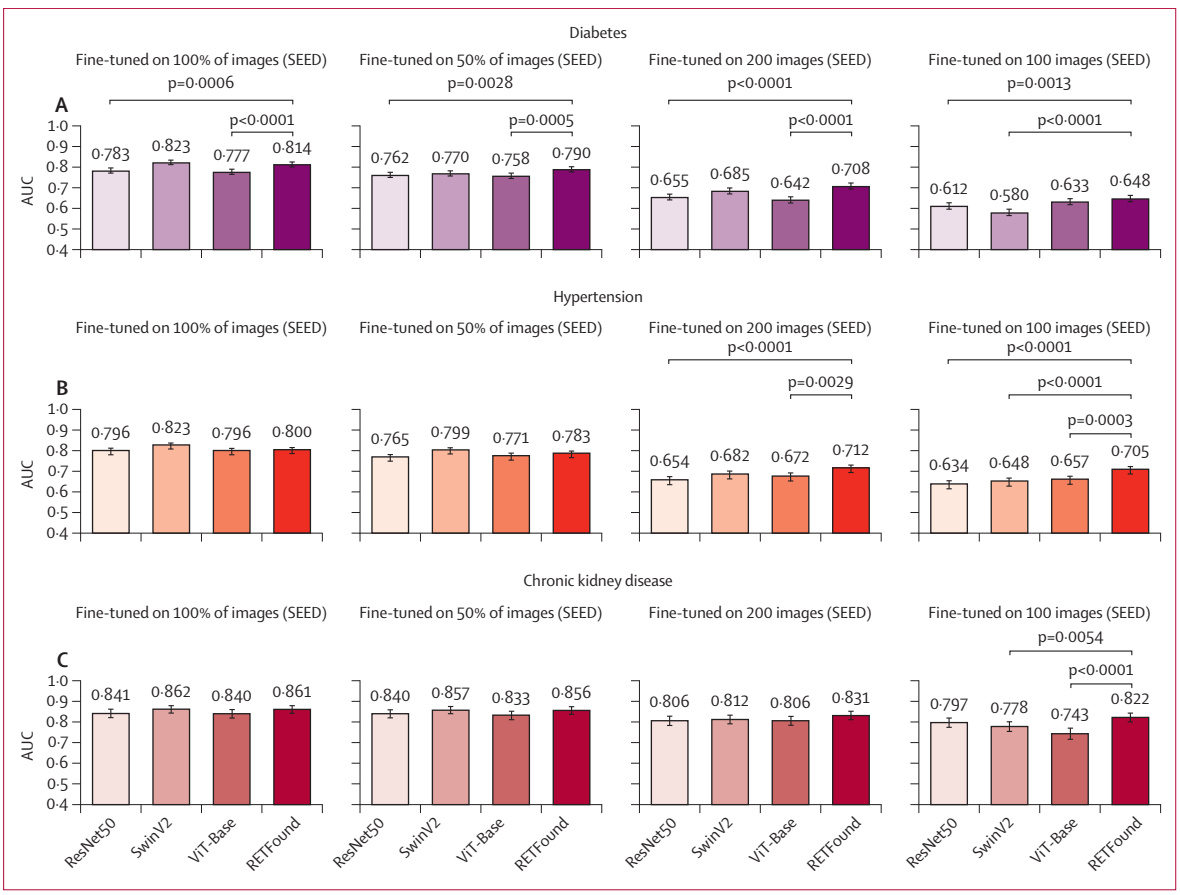


Figure 4: **Performance of the models for systemic disease detection in the internal test set when fine-tuned on various dataset sizes**
(A) Diabetes. (B) Hypertension. (C) Chronic kidney disease. Error bars are 95% CI. AUC=area under the receiver operating characteristic curve. SEED=Singapore Epidemiology of Eye Disease. Detailed pairwise p values are provided in the appendix (pp 50–51).

with previous studies showing the architectural superiority of SwinV2 over convolutional neural networks[15] and suggests that SwinV2 could serve as a competitive alternative to RETFound in low-data settings for the detection of ocular disease. Overall, findings showed that, in scenarios with abundant labelled data for ocular disease detection, traditional deep-learning model approaches could suffice to achieve the desired performance.

When fine-tuned on smaller sample sizes (≤400 images), RETFound performed better than traditional models in systemic disease detection tasks, which could be attributed to the pretraining pretext approach used in this model. RETFound used masked autoencoders on around 900 000 unlabelled retinal images,[4,45,46] which involved reconstructing highly masked retinal images and compelling the model to infer missing information from a small proportion of visible patches. This approach likely enabled RETFound to capture subtle, retina-specific contexts, such as fine-grained vascular patterns, which might be less discernible to the human eye or to traditional models fine-tuned on natural images.[42–44]

In terms of computational demands, the similar fine-tuning times for RETFound and SwinV2, despite RETFound having a larger number of parameters and therefore a greater model size, can likely be attributed to the implementation optimisations in RETFound, such as mixed-precision training and minimal data augmentation, which help to mitigate the effect of its higher parameter count. By contrast, the hierarchical architecture and shifted window mechanisms of SwinV2 introduce additional computational overhead, offsetting the efficiency typically associated with smaller models. Altogether, these factors help to explain why both models have similar training times (difference of 8 s) despite the larger parameter size of RETFound.

Across most tasks, models fine-tuned on imbalanced datasets generally achieved better performance than those trained on balanced datasets (appendix pp 5–34). One possible explanation is that the larger imbalanced datasets provided more fine-tuning data. Also, square-root inverse reweighting was applied to both balanced and imbalanced datasets; because the balanced datasets were already balanced, the effect of this reweighting was likely to be minimal there. By contrast, the reweighting applied to the imbalanced datasets could have influenced model optimisation and contributed to their better performance.

However, the present study was not designed to isolate its individual contribution.

It is important to consider the clinical relevance of these deep-learning models. For example, in the detection of diabetic retinopathy, when fine-tuned using 500 images and tested externally on the Messidor-2 dataset, there was a statistical difference ($p<0·002$) between the performances of RETFound and ResNet50 (appendix p 19). However, the difference in AUC (0·041) was clinically small. Future work on defining clinically relevant thresholds for respective tasks are warranted to better ascertain the clinical relevance of these findings.[47–49]

One strength of our study is the extensive and robust external testing on multiple datasets, both open-source and from diverse population cohort studies. Second, we included a broad range of scenarios in our evaluations by using various sample sizes for fine-tuning, both in terms of proportions of the dataset and fixed numbers of images. This approach is particularly valuable when assessing performance under different specific scenarios, including conditions of limited data availability. Third, several previous studies have fine-tuned RETFound for specific tasks, such as the detection of binary glaucoma,[50] diabetic retinopathy,[51] and abnormal or pathological OCT features[52,53] and the prediction of optic nerve metrics such as cup-to-disc ratio and retinal nerve fibre layer thickness.[54] However, none comprehensively compared traditional deep-learning models and RETFound across a range of ocular and systemic diseases and with fine-tuning on various dataset sizes. Our study is among the first to provide such a comprehensive comparison, providing a deeper understanding of the comparative strengths and limitations of each approach.

Our study also has several limitations. First, we did not include longitudinal prediction tasks and detection tasks for rare ocular and systemic diseases, which could provide deeper insights into the broader applicability of these models. Second, we did not address the environmental and economic costs of pretraining large-scale foundation models, which merit future study to balance innovation with sustainability. Third, our findings highlight the limited generalisability of current models to test sets with out-of-distribution demographics, particularly with respect to ethnicity and age. This observation is not unexpected as models generally perform better when the training and testing datasets are demographically and clinically aligned.[55] However, because our evaluation is based on the specific datasets used in this study, the extent to which these findings generalise more broadly will require future validation. Finally, we used representative standardised image resolution to align with best practices from previous studies.[11–15] Although these choices aimed to ensure stable training tailored to the architecture of each model, we acknowledge that they could introduce minor variability (eg, loss of resolution) that could potentially influence the comparisons.[56,57] Nevertheless, the promising performance of our models suggests that key diagnostic features were likely still preserved at these scales.

In terms of future work, it would first be valuable to determine whether there is an optimal minimum fine-tuning sample size and an optimal ratio of positive to negative cases to achieve reliable performance, particularly when using foundation models in data-scarce settings. The stronger baseline performance of RETFound also suggests that it could provide a more robust foundation for future AI development in retinal-based disease detection. Second, research is needed to further develop strategies such as domain adaptation and multi-institutional pretraining to improve model robustness across diverse populations. Third, future studies of emerging multimodal foundation models[5,58,59] could provide new insights and novel applications. Fourth, alternative approaches for low-data scenarios, such as low-rank adaptation[60] and few-shot class-incremental learning,[61] warrant exploration. Fifth, the current augmentation techniques were tailored differently for RETFound and the traditional deep-learning models for optimised performance. Future work could explore standardised augmentation strategies, such as a controlled ablation approach. Finally, to enhance transparency, trust, and better real-world adoption of these models among clinicians and patients, future work should focus on the interpretability and explainability of AI models.[62–66] For example, interpretable approaches such as SVIS-RULEX[62] integrate statistical, visual, and rule-based explanations to explain how deep learning-based models make decisions. In addition, explainable approaches such as GANterfactual[66] generate counterfactual images to reveal the visual features that drive classification. These frameworks represent promising directions for future exploration.

In summary, we show that traditional deep-learning models show similar performance to RETFound for ocular disease detection when fine-tuned on large, labelled datasets. However, when fine-tuned on small datasets for systemic disease detection, RETFound showed superior performance. These findings provide valuable insights and balanced perspectives into the respective merits and limitations of traditional deep-learning models and foundation models. As newer foundation models continue to emerge, further research is warranted to benchmark their performance.

Contributors

YCT, SMEY, XL, JHLG, and C-YC conceptualised and designed the study. YCT, SMEY, XL, JHLG, M-lC, MCYY, CS, YXW, JBJ, VN, ARR, CYC, and C-YC collected the data. XL, YF, JZ, RSMG, YZ, and YL developed the algorithm. YCT, SMEY, XL, JHLG, KZ, YC, SS, M-lC, KP, QH, ZDS, CCX, JBJ, GDY, YZ, PAK, YL, and C-YC analysed the data. YCT, SMEY, XL, JHLG, KZ, YC, SS, KP, QH, and GDY drafted the manuscript. All authors reviewed and approved the final manuscript. YCT and C-YC had access to the raw datasets of SEED. YCT, EST, and XS had access to the raw datasets from SP2. YXW and JBJ had access to the raw datasets from BES. MCYY had access to the raw datasets from UKBB. VN and JBJ had access to the raw datasets of CIEMS. CYC and ARR had access to the raw datasets of CUHK-STDR. YCT, SMEY, XL, JHLG, KZ, YCT, and C-YC accessed and verified all datasets used in the study. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

C-YC has acted as a consultant for Mediwhale, is a cofounder of Eye.AI, and has received travel support as a speaker from Santen. XL, RSMG, and YL are cofounders of Eye.AI. PAK has acted as a consultant for Retina Consultants of America, Topcon, Roche, Boehringer-Ingelheim, and Bitfount; is an equity owner in Big Picture Medical; has received speaker fees from Zeiss, Novartis, Gyroscope, Boehringer-Ingelheim, Apellis, Roche, AbbVie, Topcon, and Hakim Group; has received travel support from Bayer, Topcon, and Roche; has participated on advisory boards for Topcon, Bayer, Boehringer-Ingelheim, RetinAI, and Novartis; is a cofounder and equity owner of Cascader; and has received support from a UK Research & Innovation Future Leaders Fellowship and The Rubin Foundation Charitable Trust. All other authors declare no competing interests.

Data sharing

Data from the SEED, SP2, BES, CIEMS, and CHUK-STDR datasets were obtained with permission from the respective research institutes and principal investigators for use in this study only. As such, these data cannot be made publicly available. However, they can be made available upon reasonable request and with approval of the respective research institutes. To request access, please contact the corresponding authors. The UK Biobank data were obtained from the UK Biobank. Data cannot be shared publicly owing to the violation of patient privacy and lack of informed consent for data sharing. The UK Biobank dataset is available upon application through the UK Biobank access website at https://www.ukbiobank.ac.uk. The APTOS-2019 dataset is available at https://www.kaggle.com/competitions/2019-blindness-detection/data. The ODIR-5K dataset is available at https://www.kaggle.com/datasets/tanjemahamed/odir5k-classification. The PAPILA dataset is available at https://figshare.com/articles/dataset/PAPILA/14798004/1. The GAMMA dataset is available at https://huggingface.co/datasets/ctmedtech/GAMMA. The IDRiD dataset is available at https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid. The Messidor-2 dataset is available at https://www.adcis.net/en/third-party/messidor2. The source code for ResNet50, ViT-Base, and SwinV2 is publicly available in the timm (PyTorch Image Models) library, accessible at the following repository paths: ResNet50 (resnet.py, https://github.com/huggingface/pytorch-image-models/blob/main/timm/models/resnet.py), ViT-Base (vision_transformer.py, https://github.com/huggingface/pytorch-image-models/blob/main/timm/models/vision_transformer.py), and SwinV2 (swin_transformer_v2.py, https://github.com/huggingface/pytorch-image-models/blob/main/timm/models/swin_transformer_v2.py). The source code of RETFound was made publicly available by its authors and can be accessed via https://github.com/rmaphoh/RETFound_MAE.

Acknowledgments

YCT is supported by the Singapore Ministry of Health through the National Medical Research Council (NMRC) Office, MOH Holdings under the NMRC HPHSR Clinician Scientist Awards (MOH-001172-00 and MOH-001927-01) and the National University of Singapore's Presidential Young Professorship Award. C-YC is supported by the Singapore Ministry of Health through the NMRC Office, MOH Holdings under the NMRC Clinician Scientist Award – Senior Investigator (MOH-001283-00). This project is supported by A*STAR under RIE2020 Health and Biomedical Sciences Industry Alignment Fund Pre-Positioning grant number H20c6a0031. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not reflect the views of A*STAR. The authors used ChatGPT-4o for correction and proofreading of the grammar and punctuation and for enhancing conciseness. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

1 Christensen M, Vukadinovic M, Yuan N, Ouyang D. Vision–language foundation model for echocardiogram interpretation. *Nat Med* 2024; **30:** 1481–88.
2 Pai S, Bontempi D, Hadzic I, et al. Foundation model for cancer imaging biomarkers. *Nat Mach Intell* 2024; **6:** 354–67.
3 Lu MY, Chen B, Williamson DFK, et al. A visual-language foundation model for computational pathology. *Nat Med* 2024; **30:** 863–74.
4 Zhou Y, Chia MA, Wagner SK, et al. A foundation model for generalizable disease detection from retinal images. *Nature* 2023; **622:** 156–63.
5 Qiu J, Wu J, Wei H, et al. Development and validation of a multimodal multitask vision foundation model for generalist ophthalmic artificial intelligence. *NEJM AI* 2024; **1**. https://doi.org/10.1056/AIoa2300221.
6 Wang M, Lin T, Lin A, et al. Enhancing diagnostic accuracy in rare and common fundus diseases with a knowledge-rich vision-language model. *Nat Commun* 2025; **16:** 5528.
7 Bommasani R, Hudson DA, Adeli E, et al. On the opportunities and risks of foundation models. *arXiv* 2021; published online Aug 16, 2021. https://doi.org/10.48550/arXiv.2108.07258 (preprint).
8 Chia MA, Antaki F, Zhou Y, Turner AW, Lee AY, Keane PA. Foundation models in ophthalmology. *Br J Ophthalmol* 2024; **108:** 1341–48.
9 He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. 2026 IEEE Conference on Computer Vision and Pattern Recognition; June 27–30, 2016. https://doi.org/10.1109/CVPR.2016.90.
10 Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: transformers for image recognition at scale. *arXiv* 2020; published online Oct 22. https://doi.org/10.48550/arXiv.2010.11929 (preprint).
11 Yew SME, Lei X, Chen Y, et al. Deep imbalanced regression model for predicting refractive error from retinal photos. *Ophthalmol Sci* 2024; **5:** 100659.
12 Tham Y-C, Goh JHL, Anees A, et al. Detecting visually significant cataract using retinal photograph-based deep learning. *Nat Aging* 2022; **2:** 264–71.
13 Tham Y-C, Anees A, Zhang L, et al. Referral for disease-related visual impairment using retinal photograph-based deep learning: a proof-of-concept, model development study. *Lancet Digit Health* 2021; **3:** e29–40.
14 Goh JHL, Lei X, Chee M-L, et al. Multi-comparison of different ocular imaging modality-based deep learning models for visually significant cataract detection. *Ophthalmol Sci* 2025; **5:** 100837.
15 Goh JHL, Ang E, Srinivasan S, et al. Comparative analysis of vision transformers and conventional convolutional neural networks in detecting referable diabetic retinopathy. *Ophthalmol Sci* 2024; **4:** 100552.
16 Liu Z, Hu H, Lin Y, et al. Swin Transformer V2: scaling up capacity and resolution. 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition; June 18–24, 2022. https://doi.org/10.1109/CVPR52688.2022.01170.
17 Majithia S, Tham YC, Chee ML, et al. Cohort profile: the Singapore Epidemiology of Eye Diseases study (SEED). *Int J Epidemiol* 2021; **50:** 41–52.
18 Nangia V, Jonas JB, Sinha A, Matin A, Kulkarni M. Refractive error in central India: the Central India Eye and Medical Study. *Ophthalmology* 2010; **117:** 693–99.
19 Jonas JB, Xu L, Wang YX. The Beijing Eye Study. *Acta Ophthalmol* 2009; **87:** 247–61.
20 Sun Z, Tang F, Wong R, et al. OCT angiography metrics predict progression of diabetic retinopathy and development of diabetic macular edema: a prospective study. *Ophthalmology* 2019; **126:** 1675–84.
21 American Academy of Ophthalmology. Diabetic retinopathy—Europe. October, 2016. https://www.aao.org/education/topic-detail/diabetic-retinopathy-europe (accessed June 1, 2025).
22 Karthik M, Dane S. 2019 blindness detection. 2019. https://www.kaggle.com/c/aptos2019-blindness-detection (accessed Dec 1, 2025).
23 Porwal P, Pachade S, Kamble R, et al. Indian Diabetic Retinopathy Image Dataset (IDRiD). April 24, 2018. https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid (accessed Dec 1, 2025).
24 Decencière E, Zhang X, Cazuguel G, et al. Feedback on a publicly distributed image database: the Messidor database. *Image Anal Stereol* 2014; **33:** 231.

25 Ocular Disease Intelligent Recognition (ODIR-5k). 2019. https://www.kaggle.com/datasets/andrewmvd/ocular-disease-recognition-odir5k/data (accessed Dec 1, 2025).

26 Kovalyk O, Morales-Sánchez J, Verdú-Monedero R, Sellés-Navarro I, Palazón-Cabanes A, Sancho-Gómez J-L. PAPILA: dataset with fundus images and clinical data of both eyes of the same patient for glaucoma assessment. *Sci Data* 2022; **9:** 291.

27 Wu J, Fang H, Li F, et al. GAMMA challenge: glaucoma grading from multi-modality images. *Med Image Anal* 2023; **90:** 102938.

28 Tan KHX, Tan LWL, Sim X, et al. Cohort profile: the Singapore Multi-Ethnic Cohort (MEC) study. *Int J Epidemiol* 2018; **47:** 699–699j.

29 Nang EEK, Khoo CM, Tai ES, et al. Is there a clear threshold for fasting plasma glucose that differentiates between those with and without neuropathy and chronic kidney disease?: the Singapore Prospective Study Program. *Am J Epidemiol* 2009; **169:** 1454–62.

30 Chua SYL, Thomas D, Allen N, et al. Cohort profile: design and methods in the eye and vision consortium of UK Biobank. *BMJ Open* 2019; **9:** e025077.

31 Rim TH, Soh ZD, Tham YC, et al. Deep learning for automated sorting of retinal photographs. *Ophthalmol Retina* 2020; **4:** 793–800.

32 Touvron H, Cord M, Jégou H. DeiT III: revenge of the ViT. In: Avidan S, Brostow G, Cissé M, Farinella GM, Hassner T, eds. Computer vision—ECCV 2022. Lecture notes in computer science, vol 13684. Cham: Springer; 2022. https://doi.org/10.1007/978-3-031-20053-3_30.

33 Duda RO, Hart PE. Use of the Hough transformation to detect lines and curves in pictures. *Commun ACM* 1972; **15:** 11–15.

34 Cui Y, Jia M, Lin T-Y, Song Y, Belongie S. Class-balanced loss based on effective number of samples. 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition; June 15–20, 2019. https://doi.org/10.1109/CVPR.2019.00949.

35 Altalhan M, Algarni A, Turki-Hadj Alouane M. Imbalanced data problem in machine learning: a review. *IEEE Access* 2025; **13:** 13686–99.

36 Bernlohr J, Wightman R. PyTorch image models. https://github.com/huggingface/pytorch-image-models (accessed June 1, 2025).

37 PyTorch. Models and pre-trained weights. https://docs.pytorch.org/vision/stable/models (accessed June 1, 2025).

38 Tensorflow. Models. https://github.com/tensorflow/models (accessed June 1, 2025).

39 Lee AY, Yanagihara RT, Lee CS, et al. Multicenter, head-to-head, real-world validation study of seven automated artificial intelligence diabetic retinopathy screening systems. *Diabetes Care* 2021; **44:** 1168–75.

40 Hanley JA, McNeil BJ. A method of comparing the areas under receiver operating characteristic curves derived from the same cases. *Radiology* 1983; **148:** 839–43.

41 Davison AC, Hinkley DV. Tests. In: Davison AC, Hinkley DV, eds. Bootstrap methods and their application. Cambridge: Cambridge University Press, 1997: 136–90.

42 Cheung CY, Xu D, Cheng CY, et al. A deep-learning system for the assessment of cardiovascular disease risk via the measurement of retinal-vessel calibre. *Nat Biomed Eng* 2021; **5:** 498–508.

43 Rim TH, Lee CJ, Tham YC, et al. Deep-learning-based cardiovascular risk stratification using coronary artery calcium scores predicted from retinal photographs. *Lancet Digit Health* 2021; **3:** e306–16.

44 Wagner SK, Fu DJ, Faes L, et al. Insights into systemic disease through retinal imaging-based oculomics. *Transl Vis Sci Technol* 2020; **9:** 6.

45 Chen X, Xie S, He K. An empirical study of training self-supervised vision transformers. 2021 IEEE/CVF international conference on computer vision; Oct 10–17, 2021. https://doi.org/10.1109/ICCV48922.2021.00950.

46 He K, Chen X, Xie S, Li Y, Dollár P, Girshick R. Masked autoencoders are scalable vision learners. 2022 IEEE/CVF conference on computer vision and pattern recognition; June 18–24, 2022. https://doi.org/10.1109/CVPR52688.2022.01553.

47 Çorbacıoğlu ŞK, Aksel G. Receiver operating characteristic curve analysis in diagnostic accuracy studies: a guide to interpreting the area under the curve value. *Turk J Emerg Med* 2023; **23:** 195–98.

48 Willigenburg NW, Poolman RW. The difference between statistical significance and clinical relevance. The case of minimal important change, non-inferiority trials, and smallest worthwhile effect. *Injury* 2023; **54** (suppl 5)**:** 110764.

49 Mandrekar JN. Receiver operating characteristic curve in diagnostic test assessment. *J Thorac Oncol* 2010; **5:** 1315–16.

50 Chuter B, Huynh J, Hallaj S, et al. Evaluating a foundation artificial intelligence model for glaucoma detection using color fundus photographs. *Ophthalmol Sci* 2024; **5:** 100623.

51 Zhang J, Lin S, Cheng T, et al. RETFound-enhanced community-based fundus disease screening: real-world evidence and decision curve analysis. *NPJ Digit Med* 2024; **7:** 108.

52 Du K, Nair AR, Shah S, et al. Detection of disease features on retinal OCT scans using RETFound. *Bioengineering* 2024; **11:** 1186.

53 Kuo D, Gao Q, Patel D, Pajic M, Hadziahmetovic M. How foundational is the Retina Foundation Model? Estimating RETFound's label efficiency on binary classification of normal versus abnormal OCT images. *Ophthalmol Sci* 2025; **5:** 100707.

54 Chen MS, Ravindranath R, Chang R, Zhou Y, Keane PA, Wang SY. Independent evaluation of RETFound foundation model's performance on optic nerve analysis using fundus photography. *Ophthalmol Sci* 2025; **5:** 100720.

55 Zhou Y, Wang Z, Wu Y, et al. Understanding pre-training data effects in retinal foundation models using two large fundus cohorts. *Nat Commun* 2026; **17:** 3309.

56 Sabottke CF, Spieler BM. The effect of image resolution on deep learning in radiography. *Radiol Artif Intell* 2020; **2:** e190015.

57 Lakhani P. The importance of image resolution in building deep learning models for medical imaging. *Radiol Artif Intell* 2020; **2:** e190177.

58 Shi D, Zhang W, Chen X, et al. Eyefound: a multimodal generalist foundation model for ophthalmic imaging. *arXiv* 2024; published online May 18. https://doi.org/10.48550/arXiv.2405.11338 (preprint).

59 Meng W, Lin T, Lin A, et al. Enhancing diagnostic accuracy in rare and common fundus diseases with a knowledge-rich vision-language model. *Nat Commun* 2025; **16:** 5528.

60 Wang M, Lin T, Xu T, et al. Enhancing large foundation models to identify fundus diseases based on contrastive enhanced low-rank adaptation prompt. In: Bhavna A, Hao C, Fang H, Fu H, Lee CS, eds. Ophthalmic Medical Image Analysis 11th International Workshop, OMIA 2024. Nature Switzerland: Springer, 2024: 157–66.

61 Zhang J, Zhao P, Zhao Y, Li C, Hu D. Few-shot class-incremental learning for retinal disease recognition. *IEEE J Biomed Health Inform* 2025; **29:** 9001–12.

62 Ullah N, Guzmán-Aroca F, Martínez-Álvarez F, De Falco I, Sannino G. A novel explainable AI framework for medical image classification integrating statistical, visual, and rule-based methods. *Med Image Anal* 2025; **105:** 103665.

63 Singh G, Stefenon SF, Yow K-C. The shallowest transparent and interpretable deep neural network for image recognition. *Sci Rep* 2025; **15:** 13940.

64 Koh PW, Nguyen T, Tang YS, et al. Concept bottleneck models. In: Hal D III, Aarti S, eds. Proceedings of the 37th International Conference on Machine Learning. MLResearchPress, 2020: 5338–48.

65 Ribeiro MT, Singh S, Guestrin C. Why should I trust you? Explaining the predictions of any classifier. *arXiv* 2016; published online Feb 16. https://doi.org/10.48550/arXiv.1602.04938 (preprint).

66 Mertes S, Huber T, Weitz K, Heimerl A, André E. GANterfactual—counterfactual explanations for medical non-experts using generative adversarial learning. *Front Artif Intell* 2022; **5:** 825565.